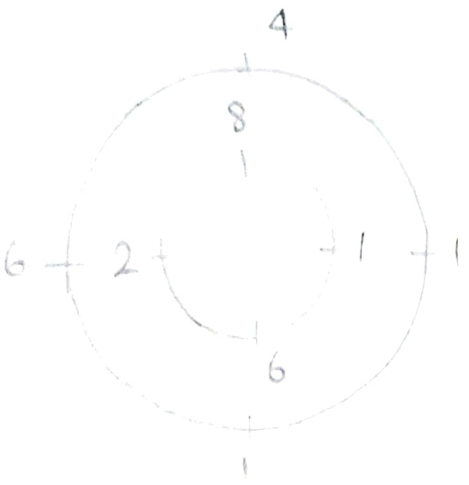
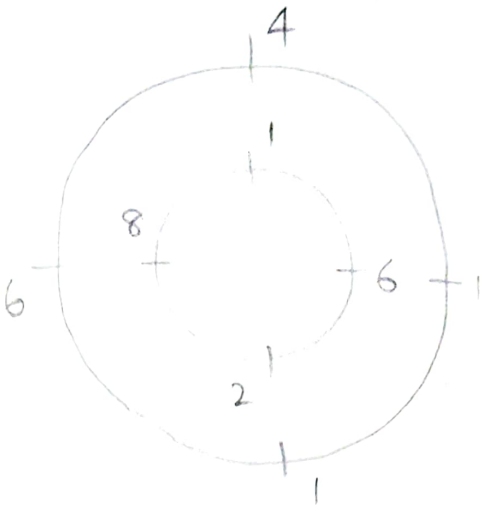
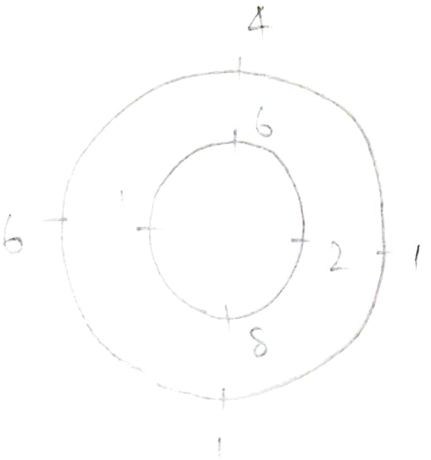
circular convolution calculation:-



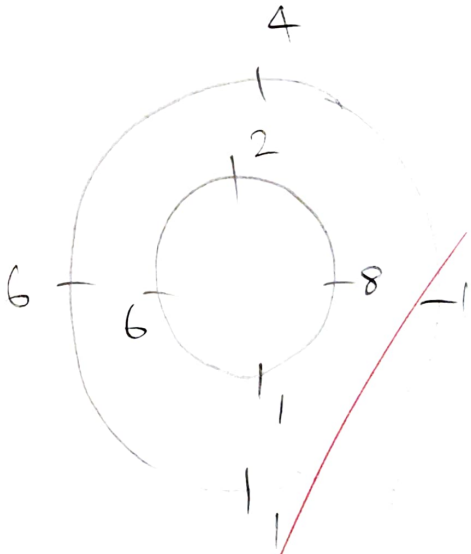
$= 1+6+12+32$
$= 51$



$= 6+2+48+4$
$= 60$



$= 2+8+6+24$
$= 40$



$= 8+8+36+1$
$= 53$

$x(n) = [51, 60, 40, 53]$

21/10/26

EXP2:- LINEAR AND CIRCULAR CONVOLUTION USING DFT

AIM:- To compute the linear and circular convolution using DFT.

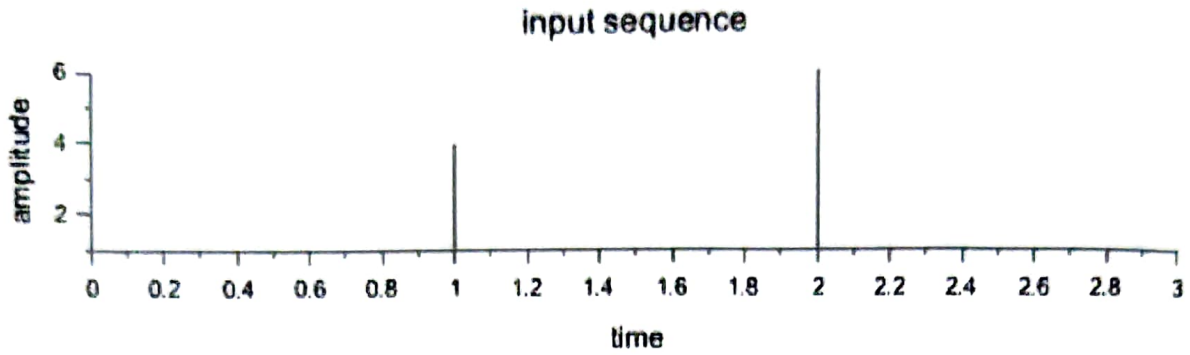
APPARATUS REQUIRED:-

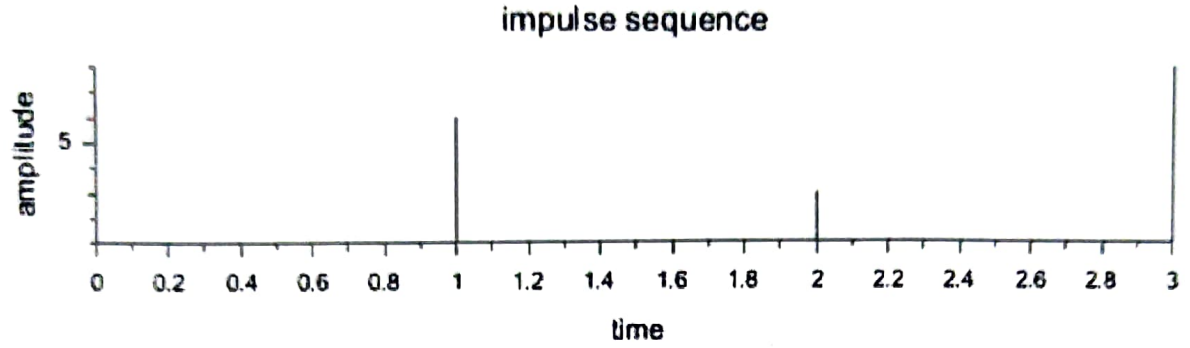
(i) Scilab Software

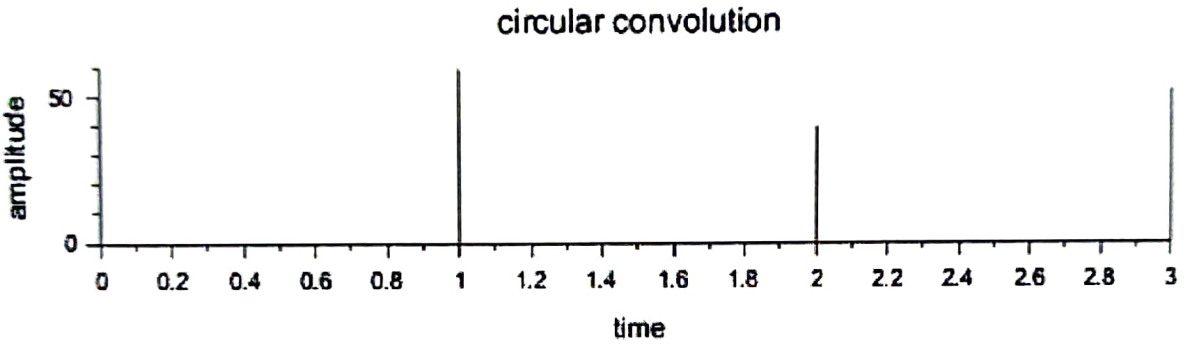
(ii) Laptop

SOURCE CODE:-

```
clc;
clear;
x=[1 4 6 1];
n1=0:1:length(x)-1;
subplot(3,1,1);
plot2d3(n1,x);
xlabel('time');
ylabel('amplitude');
title('Input sequence');
h=[1 6 2 8];
n2=0:1:length(h)-1;
subplot(3,1,2);
plot2d3(n2,h);
xlabel('time');
```


input sequence
amplitude
6
4
2
0 0.2 0.4 0.6 0.8 1 1.2 1.4 1.6 1.8 2 2.2 2.4 2.6 2.8 3
time


impulse sequence
amplitude
5
0 0.2 0.4 0.6 0.8 1 1.2 1.4 1.6 1.8 2 2.2 2.4 2.6 2.8 3
time


circular convolution
amplitude
50
0
0 0.2 0.4 0.6 0.8 1 1.2 1.4 1.6 1.8 2 2.2 2.4 2.6 2.8 3
time

```
ylabel ('amplitude');
title ('Impulse sequence');
N1 = length(x);
N2 = length(h);
N = max(N1,N2);
N3 = N1 - N2;
if (N3>0)
h = [h, zeros(1,N3)];
else
x = [x, zeros(1, abs(N3))];
end
disp (x)
disp (h)
for n=1:N
y(n)=0;
for i=1:N
j=n-i+1;
if (j<=0)
j=N+j;
end
end
disp(y)
n=0:N-1;
subplot (3,1,3);
plot2d3 (n,y);
xlabel ('time');
plot2d3 (n,y);
ylabel ('amplitude');
title ('circular convolution');
```